

VOICE-FIRST FARMING COMPANION

Kisaan AI

Practical farm intelligence,
online and off the grid.

A product case study of the Expo mobile app for smallholder farmers in Pakistan.

THE PROMISE

Weather, market prices, crop learning and AI guidance - together in the farmer's own language.



The problem is not a lack of data. It is a lack of usable decisions.

Smallholder farmers make time-sensitive choices while information is fragmented, technical, language-heavy, and often unavailable when connectivity drops.

01

Disconnected signals

Weather, crop symptoms, local prices and field context live in separate channels. Farmers must connect the dots themselves.

02

Low-confidence advice

A photo or generic answer can sound certain without acknowledging uncertainty, local conditions, or the need for verification.

03

Connectivity gaps

Live services are least reliable in the places where farm decisions are made. A useful assistant must degrade gracefully.

04

Language and literacy

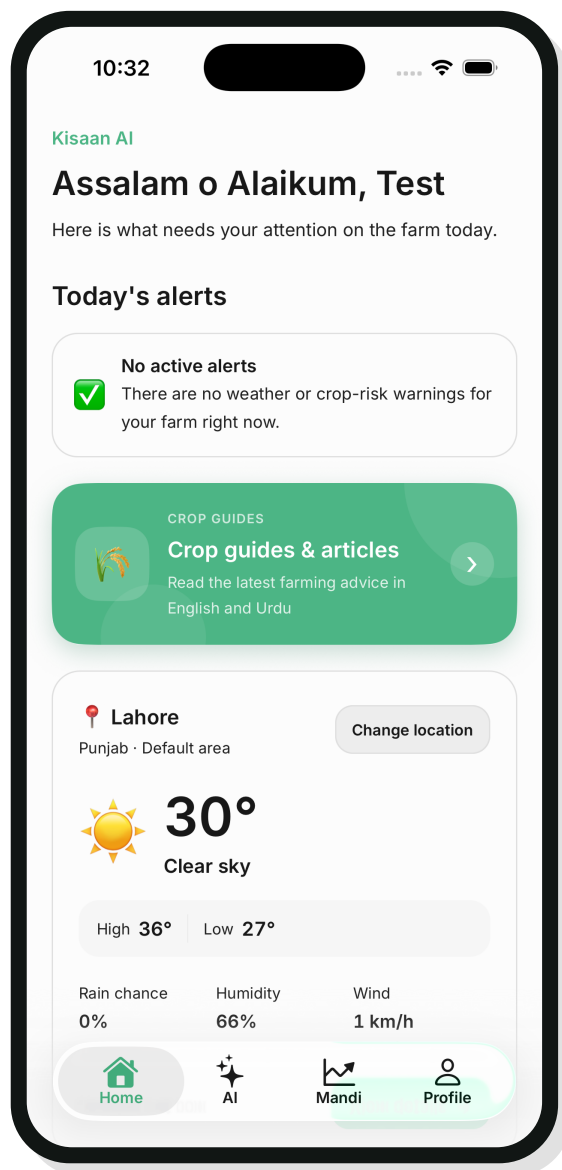
Formal English and dense interfaces create friction. Farmers need simple wording, Urdu support, readable type and spoken output.

Design challenge

How might we turn live data, local context and AI into clear next actions - without excluding farmers who prefer Urdu, speech, or have no signal?

A single decision layer for the farmer's day

Kisaan AI combines four practical jobs in one profile-available



Personalized home dashboard



Ask

Voice, text and crop photos connect the farmer to a live AI assistant with farm context.



Prepare

Local forecast, a farming advisory and risk alerts turn weather into a next action.



Sell

Daily mandi rates make price discovery searchable, bilingual and audible.



Keep working

An on-device farm helper provides text guidance after a one-time model download.

4

PRIMARY JOBS

2

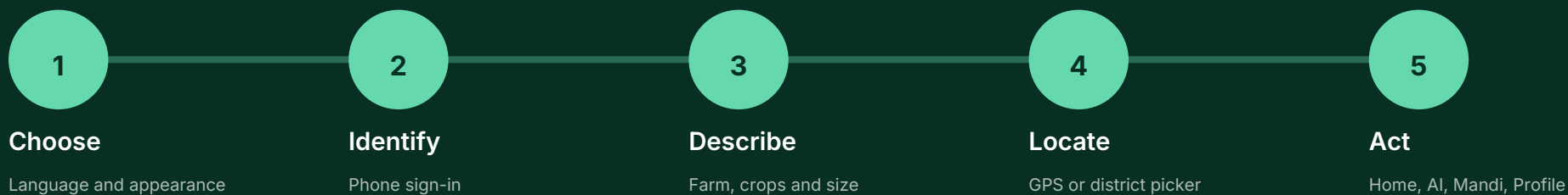
LANGUAGES

1

SHARED FARM PROFILE

From first launch to a confident next action

The experience captures only the context needed to personalize advice, then keeps the four daily jobs one tap away.



Profile as the shared context

The same crops, district, farm size and preferred language inform weather, alerts, AI prompts and settings.

Accessible interaction

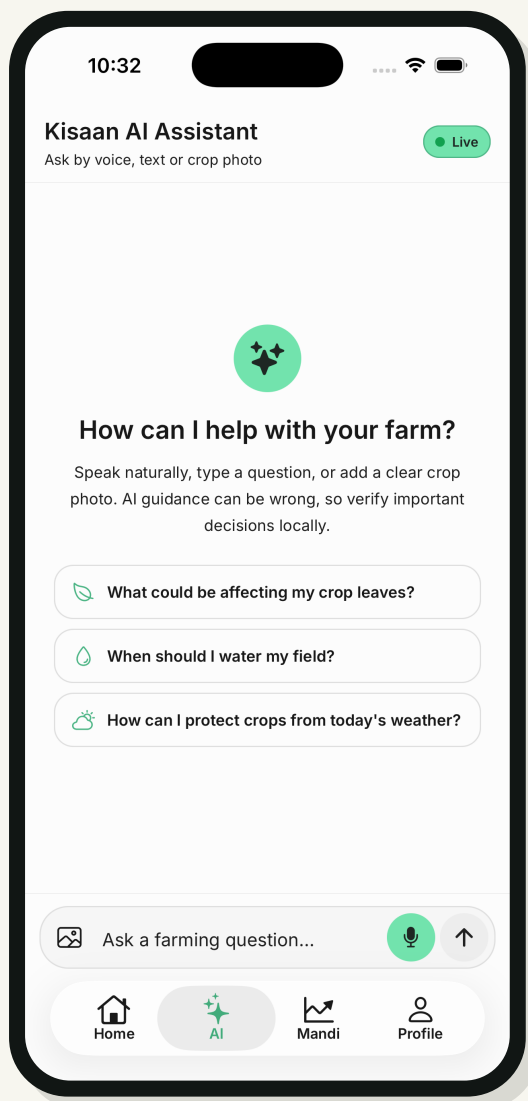
Large touch targets, bilingual labels, simple copy and text-to-speech reduce reading and navigation friction.

Resilient by design

Cached weather, prices and articles remain useful; the on-device helper covers farm Q&A without internet.

A multimodal assistant grounded in the farmer's context

Farmers can speak, type or attach a crop photo; the live session responds in simple English or everyday Urdu.



Voice, text and crop photo

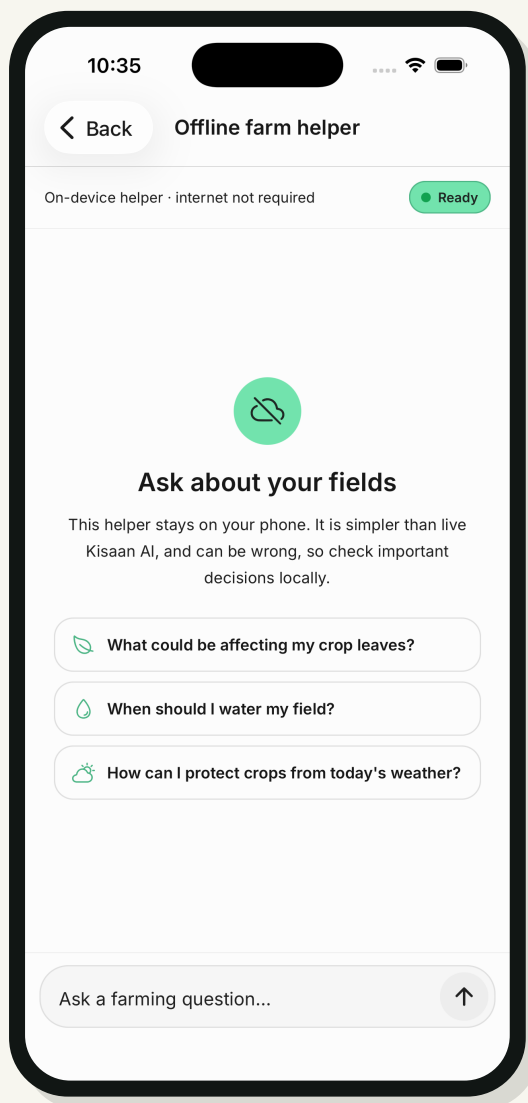
- 1 **Conversation in the natural channel**
Voice-first controls reduce typing; text and photos support quiet or visual questions.
- 2 **Context follows the farmer**
Crops, district, farm size and current app-provided mandi rates are included in the assistant context.
- 3 **Practical local language**
Responses use Pakistani units and familiar terms for irrigation, selling and field work.
- 4 **Safe starting points**
Suggested questions guide farmers toward crop leaves, watering and weather protection.

Important boundary

Live guidance can be wrong. The interface asks farmers to verify high-impact decisions locally.

Advice that stays available beyond the signal

A compact Qwen3 model runs entirely on the phone after a one-time download, keeping farm Q&A available offline.



On-device farm helper

1 No network required

After the model is stored and loaded, text questions and answers stay on the device.

2 Same farmer context

The helper uses the saved farm profile and language with a shorter prompt suited to a small model.

3 Honest capability framing

The product labels it as simpler than live Kisaan AI and repeats the need to verify advice.

4 Accessible playback

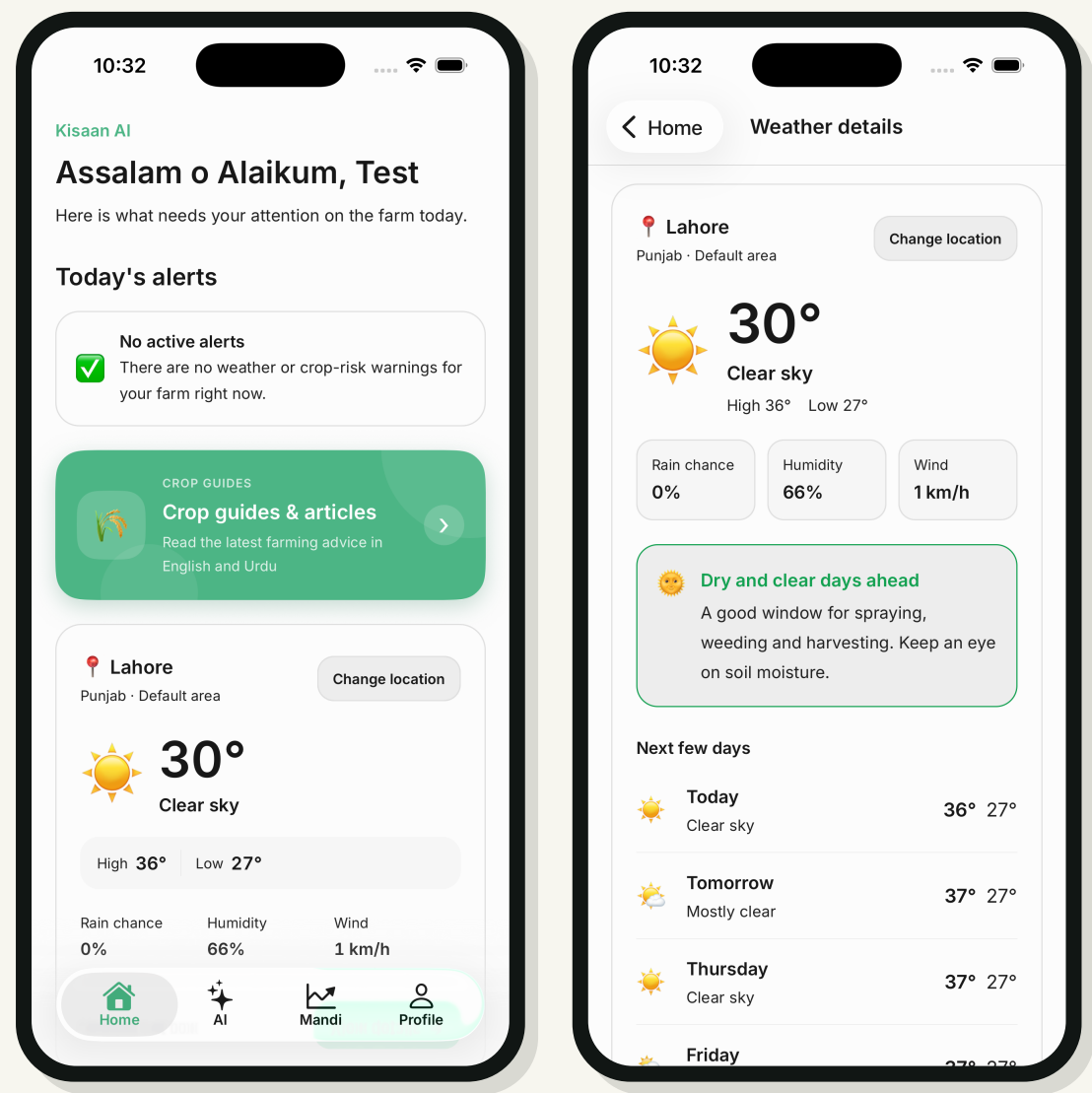
Answers can be read aloud using on-device text-to-speech.

Resilience pattern

Live AI for richer sessions; cached content and on-device inference for continuity.

Forecasts translated into farm actions

The dashboard surfaces local conditions; the detail view adds a five-day outlook and a plain-language work advisory.



Today's farm picture

Five-day forecast

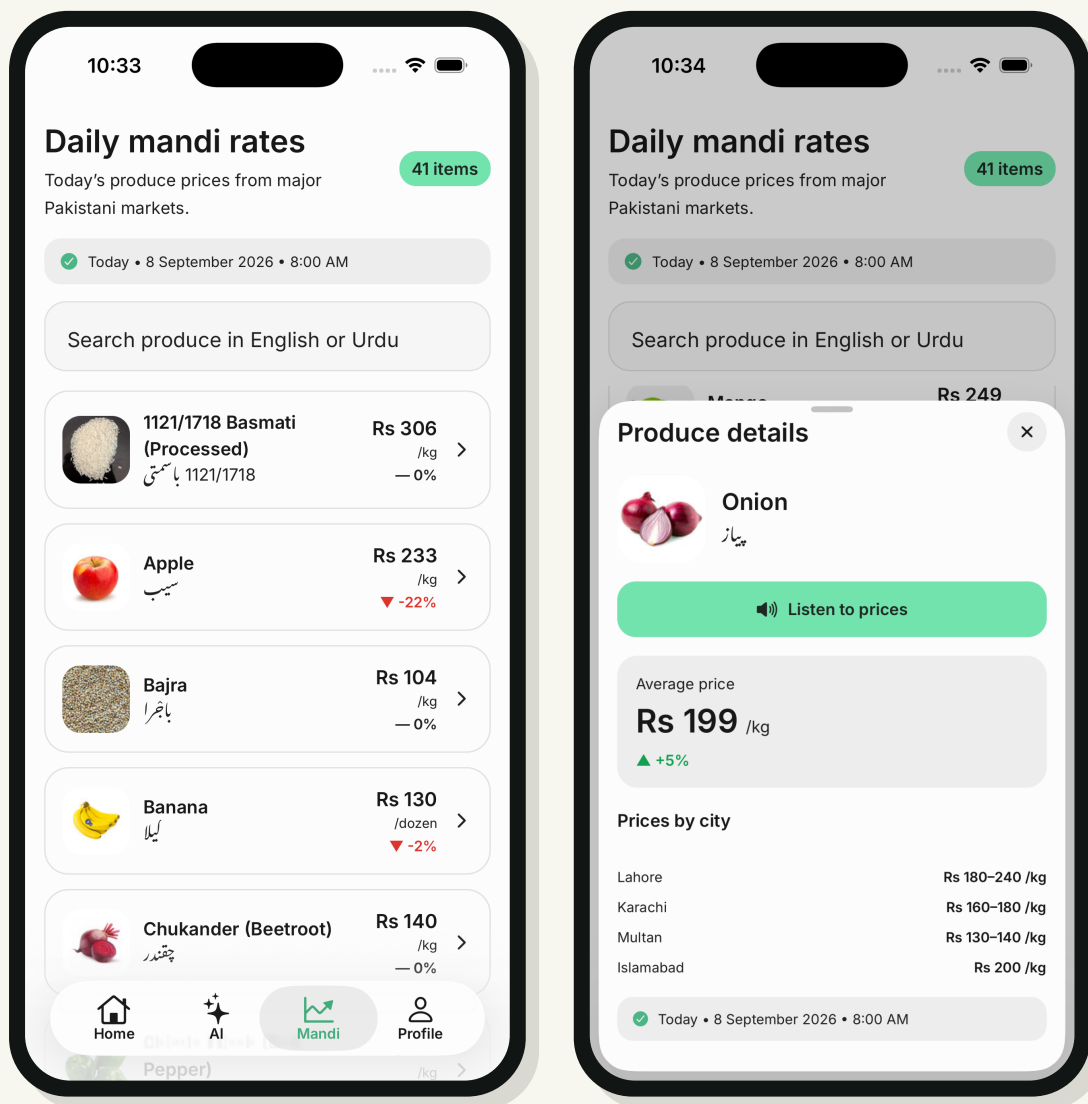
- 1 Location-aware**
Forecasts use GPS or a farmer-selected district across Pakistan's regions.
- 2 Action, not just weather**
Advisories connect dry, wet, hot, cold or windy conditions to spraying, watering and harvesting.
- 3 Prioritized risk**
Storm, rain, heat, frost, wind and wet-weather crop-risk alerts are ranked for the home screen.
- 4 Cache-aware**
A saved forecast remains visible offline and the UI communicates when the data is not fresh.

Data source

Open-Meteo powers forecasts; a shared Supabase cache reduces duplicate district fetches.

Daily mandi rates farmers can search and hear

Produce prices are presented in English and Urdu with national averages, city ranges, trend and spoken playback.



Searchable daily list

City price breakdown

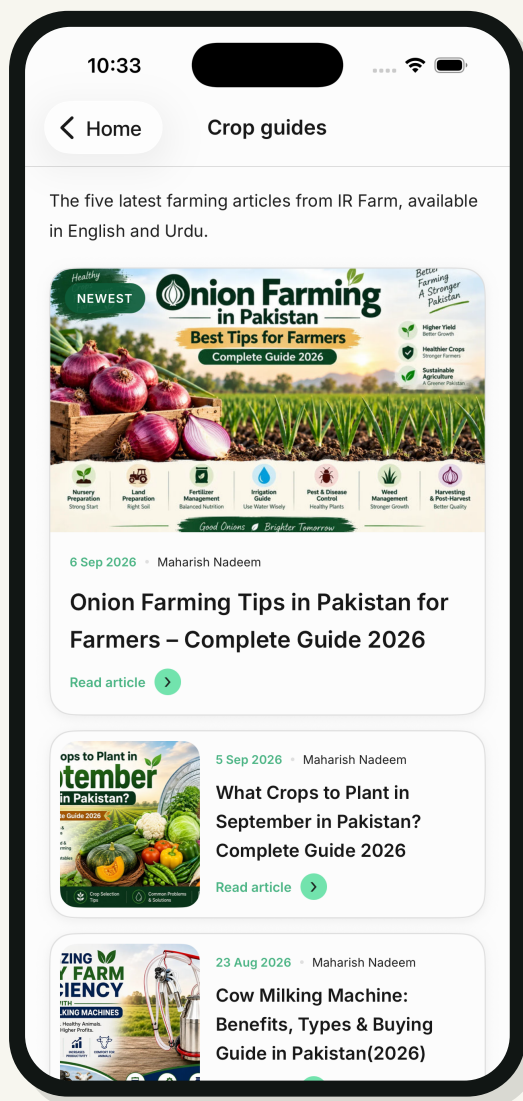
- 1 Fast price discovery**
Search works across English and Urdu names for dozens of produce items.
- 2 Useful granularity**
Each detail sheet shows an average, daily movement and city-level price ranges.
- 3 Speech playback**
A listen action makes price information available without sustained reading.
- 4 Offline continuity**
The latest device-cached list stays available when the network drops.

Current scope

Lahore, Karachi, Multan and Islamabad are shown from the daily IR Farm source.

Crop guides in both languages

Five current farming articles support deeper learning, with English and Urdu available inside each article.



Latest crop guides



Bilingual article view

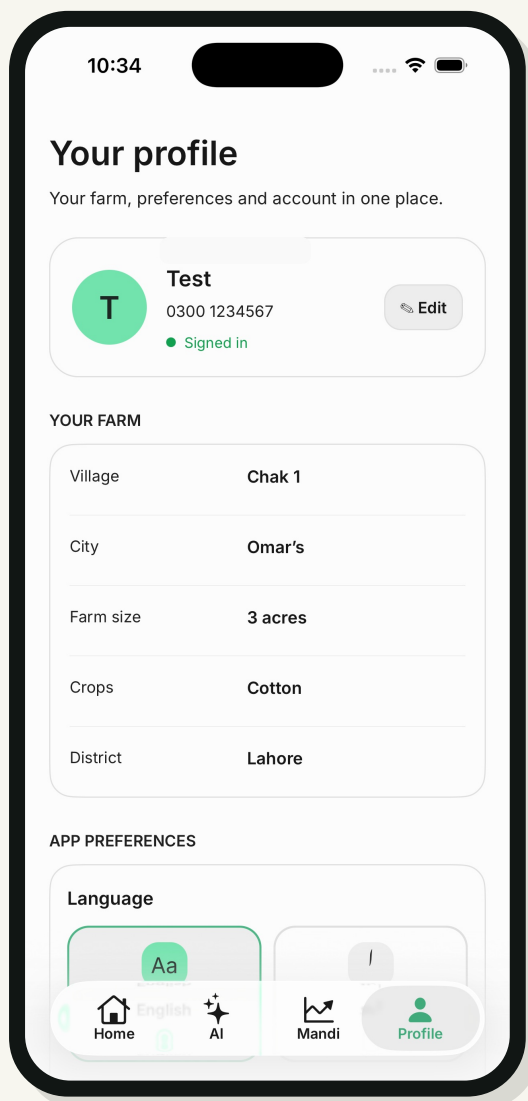
- 1 **Curated and current**
The app fetches the five latest guides and highlights the newest item.
- 2 **Bilingual reading**
A language switch keeps English and Urdu content together in one article experience.
- 3 **Visual explanation**
Illustrated cover graphics make long-form agricultural topics easier to scan.
- 4 **Offline reading**
Device caching keeps recently fetched guidance available without a connection.

Content role

Guides complement AI answers with longer, source-based educational material.

One profile, many tailored experiences

Farm details and preferences are managed in one place and reused throughout the app.



Profile and preferences

1

Farm facts

Name, village, nearest city, farm size, crops and district form the shared personalization layer.

2

Language

English and Urdu can be changed immediately, including right-to-left layouts and appropriate fonts.

3

Appearance

System, light and dark modes adapt the interface to farmer preference and environment.

4

Control

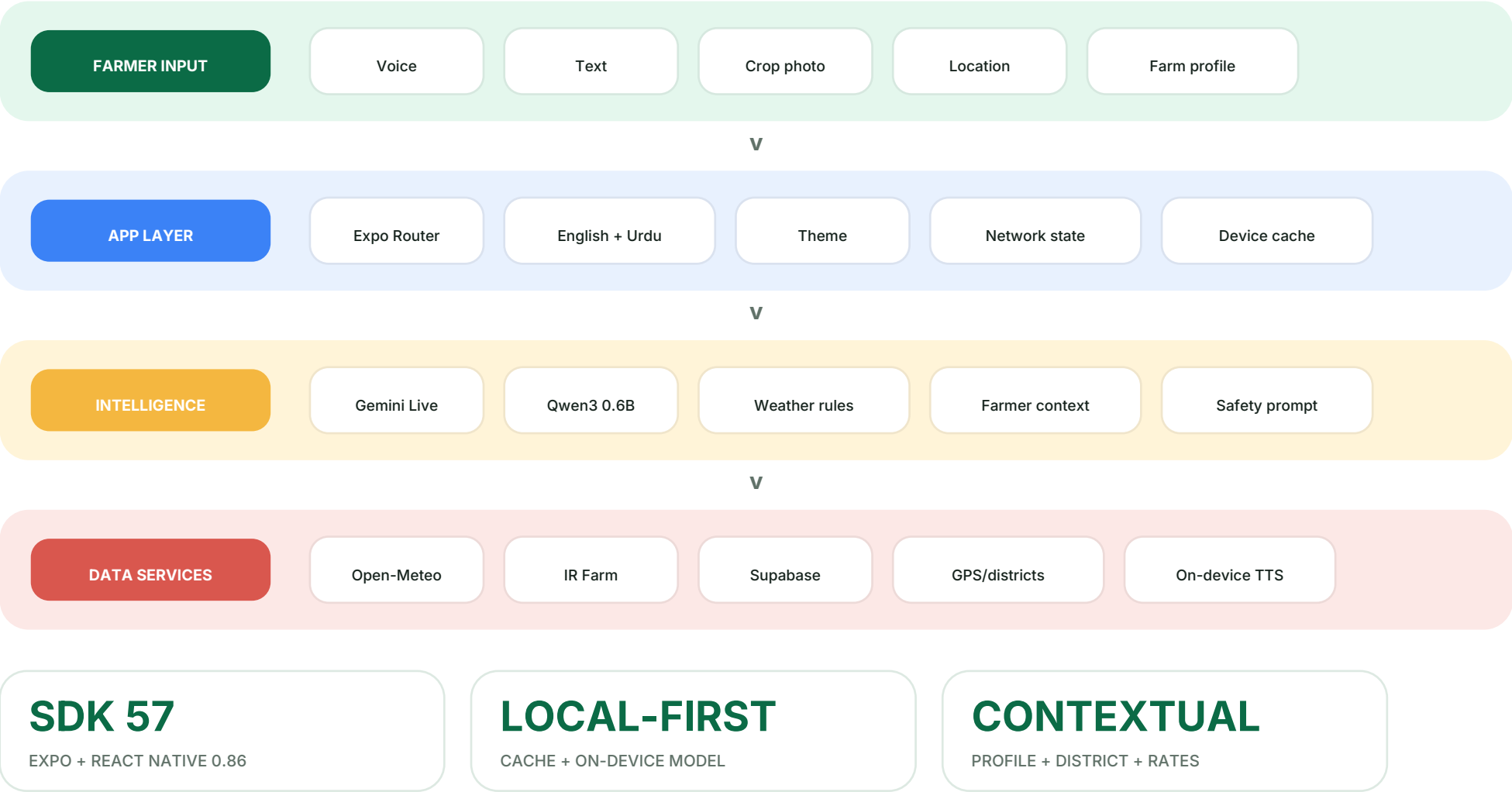
Farmers can edit details, change location, repeat onboarding or sign out.

Privacy note

The phone number in this case study screenshot is intentionally redacted.

How the product turns signals into guidance

Expo SDK 57 and React Native provide the mobile shell; providers coordinate shared farm context, network state and cached data.



Trust comes from clear boundaries

The prototype already communicates uncertainty, but production readiness depends on stronger identity, secrets and service architecture.

ALREADY DESIGNED IN

- 1 Visible uncertainty**
The AI and offline helper warn that guidance can be wrong and important decisions should be checked locally.
- 2 Careful crop-photo framing**
Guidance describes visible symptoms and offers likely causes rather than claiming a certain diagnosis.
- 3 Weather risk, not disease proof**
Wet-weather disease alerts are presented as a risk signal, not a confirmed outbreak.
- 4 Graceful degradation**
Cached content and the offline helper keep the product useful when live services fail.

BEFORE PRODUCTION

- 1 Replace demo OTP**
Current prototype OTP is generated on-device. Production must use a real SMS verification flow.
- 2 Move secrets server-side**
Gemini credentials and scraping logic should not ship as production client wiring.
- 3 Validate sources and freshness**
Show timestamps, source attribution and failure states for prices, articles and forecasts.
- 4 Field-test language and advice**
Test Urdu, speech, comprehension and agricultural usefulness with farmers and local experts.

Product principle

Kisaan AI should support judgment, not replace it. The interface must make source, freshness, uncertainty and next verification steps easy to understand.

CONCLUSION

Useful today. Resilient tomorrow.

Kisaan AI brings the farmer's most common information and advice needs into one clear mobile workflow.

- One shared farm profile personalizes every module.
- Bilingual text, voice and imagery widen access.
- Live intelligence is backed by practical offline paths.
- Weather and mandi data lead to decisions, not dashboards alone.

NORTH STAR

Help a farmer make the next good decision - wherever the field is.

